Model A · Diabetic Retinopathy · Linear Probe

Frozen RETFound-DINOv2 backbone · Linear classification head · 50 epochs

What is Linear Probing?

The backbone (ViT-Large, ~307M params) is frozen — its weights are not updated.
Only a single linear layer on top is trained to map backbone features → class scores.
This tests how much useful retinopathy information is already encoded in the pretrained RETFound representations, without any task-specific adaptation.
It is fast (~10 min) and serves as a lower-bound baseline before full fine-tuning.

Best Val AUROC	Best Epoch	Best Val κ
0.841	**38**	**0.455**

Training Configuration

Backbone	RETFound-DINOv2 ViT-Large
Pretrained on	UK fundus cohort (MEH, 736 k images)
Backbone status	FROZEN — zero gradient updates
Head	Linear layer: 1024 → 4 classes
Epochs	50 (best ckpt at epoch 30, run to 47 before power cut)
Batch size	64
Input size	224 × 224 px
Optimiser	AdamW (default RETFound settings)
Loss	CrossEntropyLoss with class weights
Class weights	R0=1.00 · R1=1.79 · R2=9.53 · R3A=15.68
Selection	Best checkpoint by val AUROC
Test eval	Not run (power cut before final epoch)

Why Class Weights?

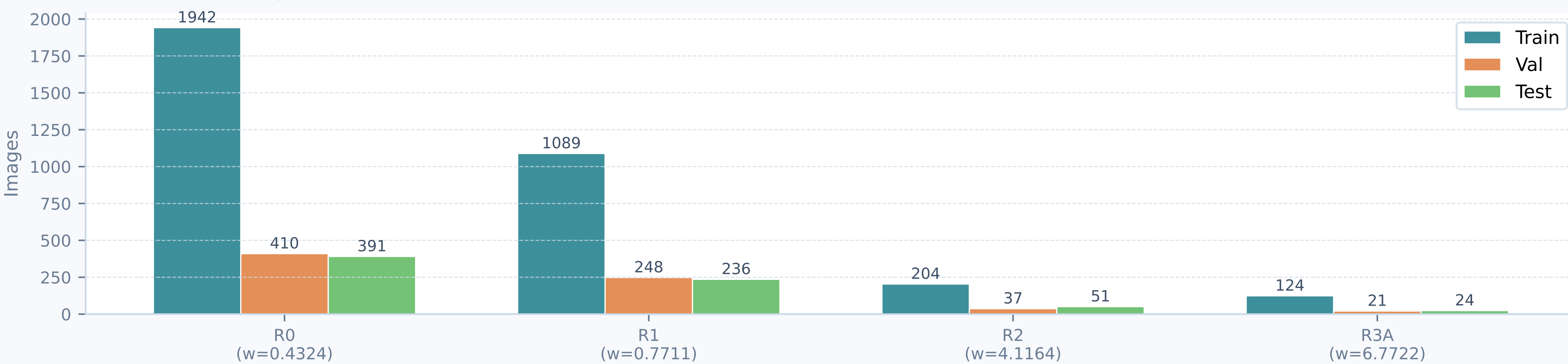
The dataset is severely imbalanced:
R0 (no retinopathy) dominates — R3A (active proliferative) is ~15× rarer.

Without weights the model learns to predict R0 for everything and achieves high accuracy while completely missing the clinically important high grades.

Weight formula: $w_i = N / (n\_classes \times count\_i)$
then scaled so the minimum weight = 1.0.

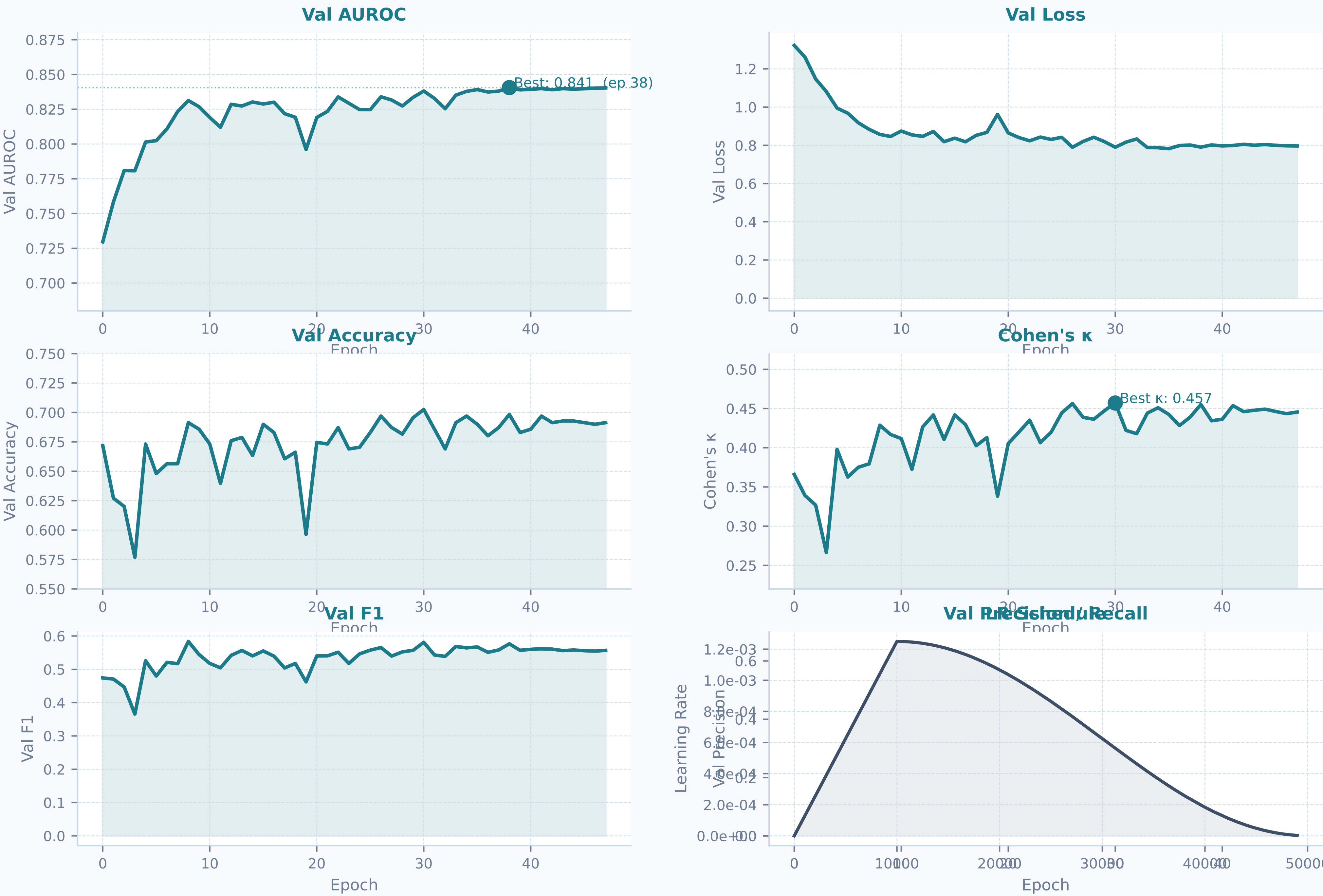
Effect: each class contributes equally to the loss regardless of frequency, forcing the model to pay attention to rare but critical R2 and R3A cases.

Dataset · Image counts per class and split



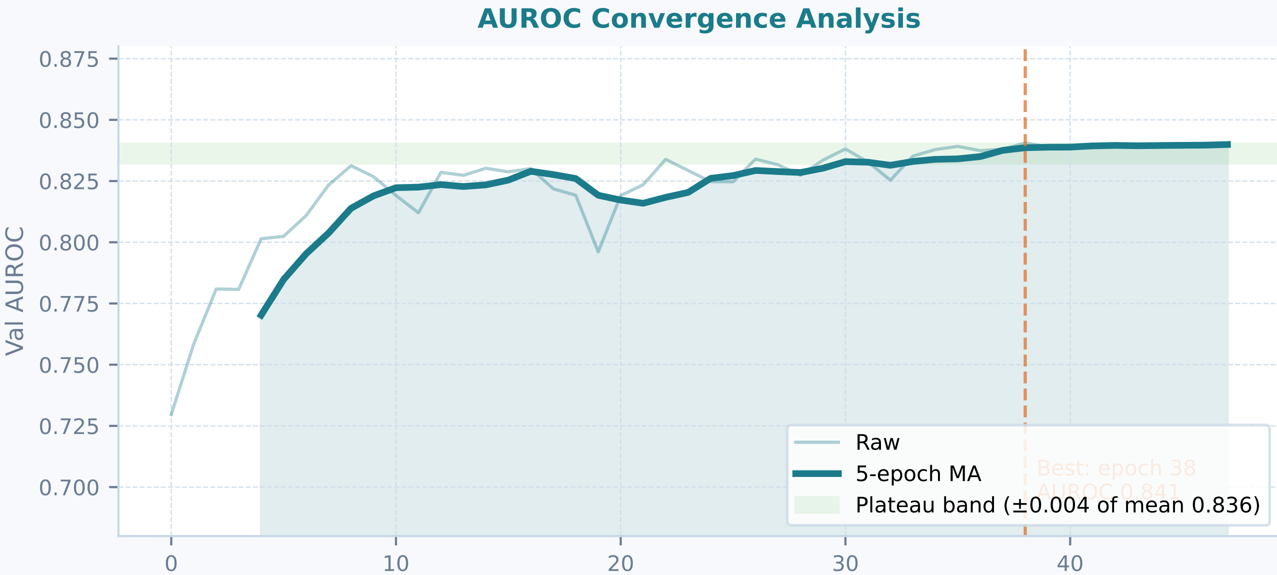
Training Dynamics — Validation Metrics over Epochs

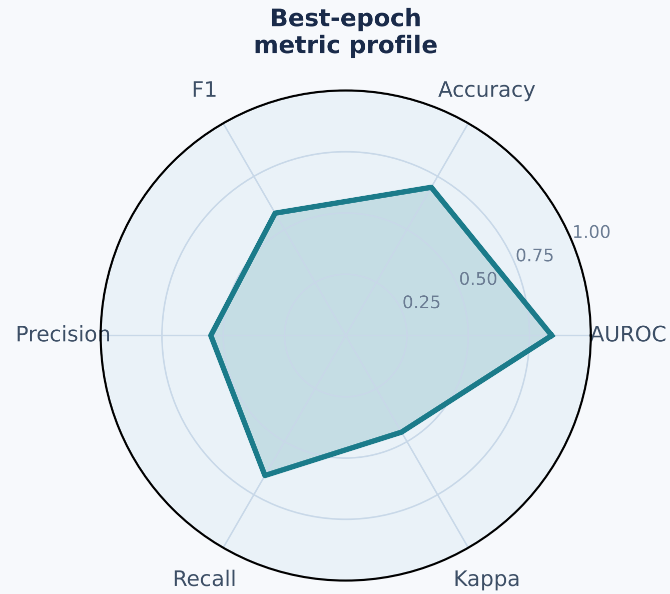
Backbone frozen throughout · Only the linear head is updated each epoch



Analysis & Interpretation

Convergence behaviour · metric interpretation · limitations





Validation Metrics — Selected Epochs

Epoch	AUROC	Accuracy	F1	Kappa	Precision	Recall
0	0.7296	0.6718	0.4742	0.3660	0.6645	0.4517
4	0.8014	0.6732	0.5258	0.3980	0.5241	0.5940
8	0.8313	0.6913	0.5838	0.4287	0.5973	0.6354
38 ★	0.8406	0.6983	0.5765	0.4553	0.5506	0.6597
47	0.8403	0.6913	0.5570	0.4455	0.5333	0.6438

Key Observations

- AUROC rose rapidly in the first 8 epochs (0.73 → 0.83), showing RETFound features already encode retinopathy-relevant structure.
- The model plateaued around 0.838–0.840 from epoch 25 onwards — the linear head had saturated; more training offered diminishing returns.
- Val accuracy (~70%) and κ (~0.46) are modest because a single linear boundary cannot perfectly separate 4 overlapping ordinal classes.
- No test set evaluation was obtained (power cut at epoch 48, best checkpoint at epoch 30 was preserved).
- This baseline establishes a lower bound: AUROC 0.840 from a frozen backbone with only a linear head.